Assignment-02

Design and Implementation of a Multi-Step Prompt Workflow for
Personalized Study Plan Generation

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1. Introduction

Students often need a study plan that matches their available time, current knowledge, subjects, and examination goals. A single broad prompt may produce a timetable that is too demanding, omit difficult topics, or fail to include revision and practice. A structured workflow can divide this task into smaller stages and check the plan before it is used.

This assignment presents a multi-step prompt workflow for personalized study plan generation. The process contains three connected stages: learner analysis and planning, learning-material generation, and quality review. Each stage uses the output of the previous stage so the final plan remains organized and relevant.

2. Objective

The main objective is to design a reliable prompt workflow that converts a learner profile into a practical study plan. The workflow aims to:

• Identify subjects, strengths, weak areas, deadlines, and available study hours.
• Arrange topics from foundational concepts to advanced material.
• Create a realistic schedule with practice and revision activities.
• Keep generated material aligned with the approved plan.
• Review the plan for missing topics, overloaded days, and unclear tasks.

3. Workflow Design

The workflow is organized as a pipeline of three specialized stages. Stage 1 analyzes the learner and creates a plan. Stage 2 creates daily tasks and practice material based on that plan. Stage 3 checks the output for coverage, feasibility, and consistency before producing the final version.

Stage 1 – Planning: Defines the learner profile, subjects, priorities, time limits, and milestones.

Stage 2 – Drafting: Generates a daily schedule, short explanations, exercises, and revision sessions.

Stage 3 – Review: Finds missing topics, unrealistic workloads, repetition, and formatting issues.

This separation makes it easier to revise the plan before investing effort in detailed learning material.

4. Methodology

The workflow was developed using three chained prompts. The output of each prompt becomes the input or reference for the next stage. This reduces the chance of producing a schedule that ignores the learner's actual constraints.

4.1 Step 1: Learner Profile and Study Plan

The first stage gathers the learner's current level, subjects, target date, available hours, and difficult topics. It then proposes a sequence and weekly milestones.

> Prompt 1: Act as an experienced academic study coach. Analyze the learner profile below. Identify the subjects, current skill level, weak areas, target date, available study hours, and three to five learning objectives. Create a realistic weekly study outline with priorities and revision milestones. Do not generate the full daily lessons yet. Learner Profile: [Insert Profile] Subjects: [Insert Subjects] Target Date: [Insert Date].

4.2 Step 2: Daily Learning Material Generation

The second stage uses the approved outline to produce a practical schedule. Each day should contain specific tasks, estimated time, and a short practice activity. The prompt limits the model to the approved subjects and time budget.

> Prompt 2: Using only the approved study plan from Step 1, prepare a daily schedule. For each session, include the topic, estimated duration, one learning activity, and one practice question or task. Include short revision sessions and a weekly self-check. Keep the workload within the learner's available hours and do not introduce unrelated subjects.

4.3 Step 3: Review, Validation and Refinement

The third stage acts as a quality gate. It checks whether the schedule is complete and realistic before presenting it to the learner.

> Prompt 3: Act as a strict academic planner. Review the study schedule using this checklist: (1) Are all learning objectives covered? (2) Does each day fit the available time? (3) Are difficult topics given enough practice? (4) Are revision and self-checks included? (5) Are tasks clear and non-repetitive? (6) Does the schedule follow the requested format? List issues first, then provide the corrected final study plan.

5. Results and Discussion

The multi-step workflow provides a systematic way to produce personalized study plans. The planning stage identifies priorities and constraints before the schedule is created. This gives the drafting stage a clear structure and reduces the likelihood of leaving out important subjects.

The second advantage is controlled generation. Because daily tasks are based on an approved plan and a defined time budget, the schedule can remain practical and focused. The learner can also change priorities at the planning stage without rewriting every part of the output.

The review stage provides an additional safeguard. It can identify overloaded days, missing revision, repeated activities, and tasks that do not support the stated objectives. Correcting these issues makes the final schedule easier to follow.

The workflow does not guarantee academic results; its usefulness depends on the accuracy of the learner profile and the quality of the review. However, dividing the task into stages makes the process easier to evaluate and improve.

6. Conclusion

The design of a multi-step prompt workflow demonstrates how prompt engineering can support personalized study planning. Rather than relying on one overloaded prompt, the approach separates learner analysis, schedule generation, and quality review into three connected stages.

Each stage creates a checkpoint for validating goals, time constraints, learning activities, and final presentation. The same workflow can be adapted to different subjects, exam dates, skill levels, and study preferences. Therefore, a multi-step prompt workflow is a practical and repeatable method for generating more consistent study plans.